AI23331 Fundamentals of Machine Learning

A PROJECT REPORT

AI Powered Crop Disease Detection System Using CNN

Submitted by

2116230701241 -PRAVEEN KUMAR D

2116230701056 -BOOTHALINGESH N

May, 2026

BONAFIDE CERTIFICATE

Certified that this project report "AI Powered Crop Disease Detection System Using CNN" is the bonafide work of "PRAVEEN KUMAR D(230701241), BOOTHALINGESH N(230701056)" who carried out the project work under my supervision.

SIGNATURE OF THE FACULTY INCHARGE

Submitted for the Practical Examination held on _____

SIGNATURE OF THE INTERNAL EXAMINER

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	
1	INTRODUCTION	3
2	LITERATURE REVIEW	4
3	PROPOSED SYSTEM	5
	3.1 ARCHITECTURE DIAGRAM	
4	MODULES DESCRIPTION	7
	4.1 MODULE 1	
	4.2 MODULE 2	
5	IMPLEMENTATION AND RESULTS	
	5.1 EXPERIMENTAL SETUP	9
	5.2 RESULTS	
6	CONCLUSION AND FUTURE WORK	12
7	REFERENCES	13

ABSTRACT

This project presents the development of an AI-powered Crop Disease Detection System using Convolutional Neural Networks (CNN) and Deep Learning techniques. The system is designed to identify crop diseases from leaf images and provide accurate predictions through a web-based application. TensorFlow and Keras are used to train the CNN model on a dataset containing healthy and diseased crop images. Image preprocessing techniques such as rescaling, rotation, zooming, and flipping are applied to improve model accuracy and performance.

The application is developed using Flask for the backend and HTML, CSS, and JavaScript for the frontend interface. Users can upload crop leaf images, and the system predicts the disease along with the confidence score in real time. The project also generates evaluation metrics such as accuracy graphs, loss graphs, confusion matrix, and classification reports. This system demonstrates the practical use of Artificial Intelligence in smart agriculture for early disease detection and crop management.

The proposed system is efficient, scalable, and capable of supporting future enhancements such as mobile integration, cloud deployment, and advanced deep learning architectures. This system highlights the importance of AI-based technologies in improving agricultural productivity and enabling smart disease management solutions for modern farming.

1. INTRODUCTION

Agriculture is one of the most important sectors in the world, playing a major role in food production and economic development. Farmers depend heavily on healthy crops for better productivity and profit. However, crop diseases are one of the major challenges faced in agriculture, as they can reduce crop quality, decrease yield, and cause significant financial losses. Many plant diseases spread rapidly and affect large agricultural areas if they are not identified at an early stage. Traditional disease detection methods mainly depend on manual observation by agricultural experts, which can be time-consuming, expensive, and sometimes inaccurate. In many rural areas, farmers may not have easy access to agricultural specialists, making early disease detection difficult.

With the advancement of Artificial Intelligence (AI) and Deep Learning technologies, automated crop disease detection systems have become more efficient and reliable. Deep Learning models, especially Convolutional Neural Networks (CNN), have shown excellent performance in image classification and object recognition tasks. CNN models can automatically learn important features from images and accurately classify different crop diseases. These technologies provide an effective solution for detecting plant diseases quickly and accurately without requiring expert supervision.

This project focuses on the development of an AI-powered Crop Disease Detection System using Convolutional Neural Networks and Deep Learning techniques. The main objective of the project is to identify crop diseases from leaf images and provide real-time predictions through a web-based application. The system is trained using a dataset containing images of healthy and diseased crop leaves. During training, the CNN model learns the unique patterns, colors, and textures associated with different plant diseases. Once trained, the

model can predict the disease category when a user uploads a crop image.

The system uses TensorFlow and Keras frameworks for developing and training the CNN model. Image preprocessing techniques such as image rescaling, rotation, zooming, and horizontal flipping are applied to improve model performance and accuracy. These preprocessing techniques help the model generalize better and reduce overfitting. The trained model is capable of analyzing crop leaf images and generating predictions with confidence scores in real time.

The backend of the application is developed using the Flask web framework, which manages image uploads, prediction processing, and communication between the user interface and the AI model. The frontend of the application is designed using HTML, CSS, and JavaScript to provide a modern, responsive, and user-friendly environment. Users can upload crop images through the web interface, preview the selected image, and receive disease prediction results instantly. The application also includes drag-and-drop image upload functionality for better user experience.

In addition to disease prediction, the project also evaluates the performance of the CNN model using various performance metrics such as accuracy graphs, loss graphs, confusion matrix, and classification reports. These evaluation techniques help measure the effectiveness of the model and analyze prediction performance. Early stopping techniques are also used during training to prevent overfitting and improve the overall efficiency of the model.

This project highlights the importance of AI-based technologies in modern agriculture and smart farming. In the future, the system can be further enhanced by integrating mobile applications, cloud deployment, multilingual support, and advanced deep learning models such as ResNet or MobileNet.

2. LITERATURE REVIEW

Crop disease detection has become an important research area in modern agriculture due to the increasing demand for improved crop productivity and food security. Traditional methods of plant disease identification mainly depend on visual inspection by agricultural experts. Although manual detection methods can provide accurate results, they are time-consuming, expensive, and not always accessible to farmers in remote areas. To overcome these limitations, researchers have explored the use of Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning techniques for automatic crop disease detection.

Early research in plant disease detection focused on image processing techniques such as color analysis, texture extraction, and shape detection. These methods used handcrafted features to identify diseases from leaf images. However, traditional image processing methods often failed to achieve high accuracy when dealing with complex disease patterns, varying lighting conditions, and large datasets. As a result, researchers shifted toward Machine Learning approaches, where classification algorithms such as Support Vector Machine (SVM), Decision Tree, and Random Forest were used for disease identification. These algorithms improved prediction performance compared to traditional methods but still required manual feature extraction, which limited their efficiency and scalability.

Recent advancements in Deep Learning have significantly improved the performance of image classification systems. Convolutional Neural Networks (CNN) have become one of the most widely used Deep Learning models for plant disease detection because of their ability to automatically extract important features from images. CNN models can identify complex patterns, textures, and color variations directly from raw image data without

requiring manual feature engineering. Several studies have demonstrated that CNN-based models achieve higher accuracy than traditional Machine Learning techniques in crop disease classification tasks.

Researchers have successfully applied Deep Learning frameworks such as TensorFlow, Keras, and PyTorch for building plant disease detection systems. Publicly available datasets like PlantVillage have been widely used for training and testing crop disease classification models. Studies have shown that data augmentation techniques such as image rotation, flipping, zooming, and rescaling help improve model performance and reduce overfitting. These preprocessing methods increase dataset diversity and improve the generalization capability of CNN models.

Many research works have also focused on deploying crop disease detection systems through web and mobile applications to make them more accessible to farmers. Flask and Django frameworks are commonly used to integrate trained Deep Learning models into web applications. These systems allow users to upload crop leaf images and receive disease predictions in real time. Mobile-based disease detection applications have also gained popularity due to the increasing use of smartphones in agriculture.

Performance evaluation plays an important role in crop disease detection systems. Researchers commonly use metrics such as accuracy, precision, recall, F1-score, confusion matrix, and classification reports to measure model effectiveness. Several studies have reported high accuracy rates above 90% using CNN architectures for crop disease classification. Advanced CNN models such as ResNet, MobileNet, Inception, and VGGNet have also been explored to further improve disease prediction accuracy and computational efficiency.

Recent advancements in Artificial Intelligence and computer vision technologies have further improved the efficiency of crop disease detection systems. Researchers have explored transfer learning techniques using pre-trained deep learning models such as VGG16, ResNet, InceptionV3, and MobileNet for plant disease classification. These models reduce training time and improve prediction accuracy, especially when working with limited datasets.
Studies have shown that transfer learning models can achieve highly accurate results in identifying multiple crop diseases under different environmental conditions.

Researchers have also focused on real-time disease detection systems using mobile devices and cloud computing technologies. Smartphone-based applications allow farmers to capture crop images directly from fields and receive instant disease predictions. Cloud-based deployment enables large-scale accessibility and remote monitoring of crop health. These technologies make AI-powered disease detection systems more practical and accessible for farmers in rural and remote areas.

Several studies have highlighted the importance of dataset quality and preprocessing in improving model performance. Variations in lighting conditions, background noise, image resolution, and leaf orientation can affect prediction accuracy. To address these challenges, preprocessing techniques such as image normalization, segmentation, noise removal, and augmentation are commonly used. Data augmentation increases dataset diversity and helps CNN models generalize better on unseen images.

In addition, researchers have explored integrating crop disease detection systems with Internet of Things (IoT) devices and sensor technologies for smart farming applications. IoT-based monitoring systems can collect environmental data such as temperature, humidity, and soil moisture, which can further improve disease prediction accuracy. These intelligent systems support precision agriculture by enabling continuous crop monitoring and early disease management. Overall, the literature indicates that AI-based crop disease detection

3. PROPOSED SYSTEM

The proposed system is an AI-powered Crop Disease Detection System designed to identify plant diseases from crop leaf images using Deep Learning and Convolutional Neural Networks (CNN). The system provides an automated and efficient solution for detecting diseases at an early stage, helping farmers reduce crop damage and improve agricultural productivity. The application is developed as a web-based platform where users can upload crop leaf images and receive disease prediction results instantly.

The system consists of four major components: Image Preprocessing Layer, CNN Model Training Layer, Flask Backend Layer, and Web Application Interface Layer. These components work together to create a complete pipeline from image upload to disease prediction and result display.

The first stage of the system is image preprocessing. Crop leaf images collected from the dataset are resized and normalized before being used for training. To improve model performance and reduce overfitting, image augmentation techniques such as rotation, zooming, horizontal flipping, and rescaling are applied using TensorFlow's ImageDataGenerator. These preprocessing techniques increase dataset diversity and help the CNN model generalize better for different image conditions.

The second stage involves training the CNN model using TensorFlow and Keras frameworks. The Convolutional Neural Network consists of multiple convolutional layers, pooling layers, dense layers, and dropout layers. The convolutional layers extract important features such as texture, shape, and color patterns from leaf images, while pooling layers reduce dimensional complexity. The dense layers perform classification and predict the disease category. The model is trained using labeled crop disease datasets, and early stopping techniques are applied to improve efficiency and avoid overtraining.

The third stage is backend integration using the Flask web framework. Flask acts as the communication bridge between the frontend interface and the trained CNN model. When a user

uploads a crop image through the website, the Flask backend processes the image and sends it to the CNN model for prediction. The model analyzes the image and returns the predicted disease along with the confidence score.

The frontend interface is developed using HTML, CSS, and JavaScript to provide a modern and user-friendly experience. Users can upload images through drag-and-drop functionality or file selection options. The system displays the uploaded image, predicted disease name, and confidence percentage in real time.

The proposed system also includes performance evaluation features such as accuracy graphs, loss graphs, confusion matrix visualization, and classification reports to measure the effectiveness of the CNN model. Overall, the system provides a fast, accurate, and intelligent solution for crop disease detection and demonstrates the practical application of Artificial Intelligence in smart agriculture and modern farming technologies.

3.1 Architecture Diagram

The architecture of the proposed system follows a simple and efficient flow for crop disease detection. The user first uploads a crop leaf image through the web interface developed using HTML, CSS, and JavaScript. The uploaded image is sent to the Flask backend server for processing and validation. The backend then passes the image to the trained CNN deep learning model for disease prediction. The CNN model analyzes the image features and identifies the disease category along with the confidence score. Finally, the prediction result is displayed to the user through the web application interface in real time.

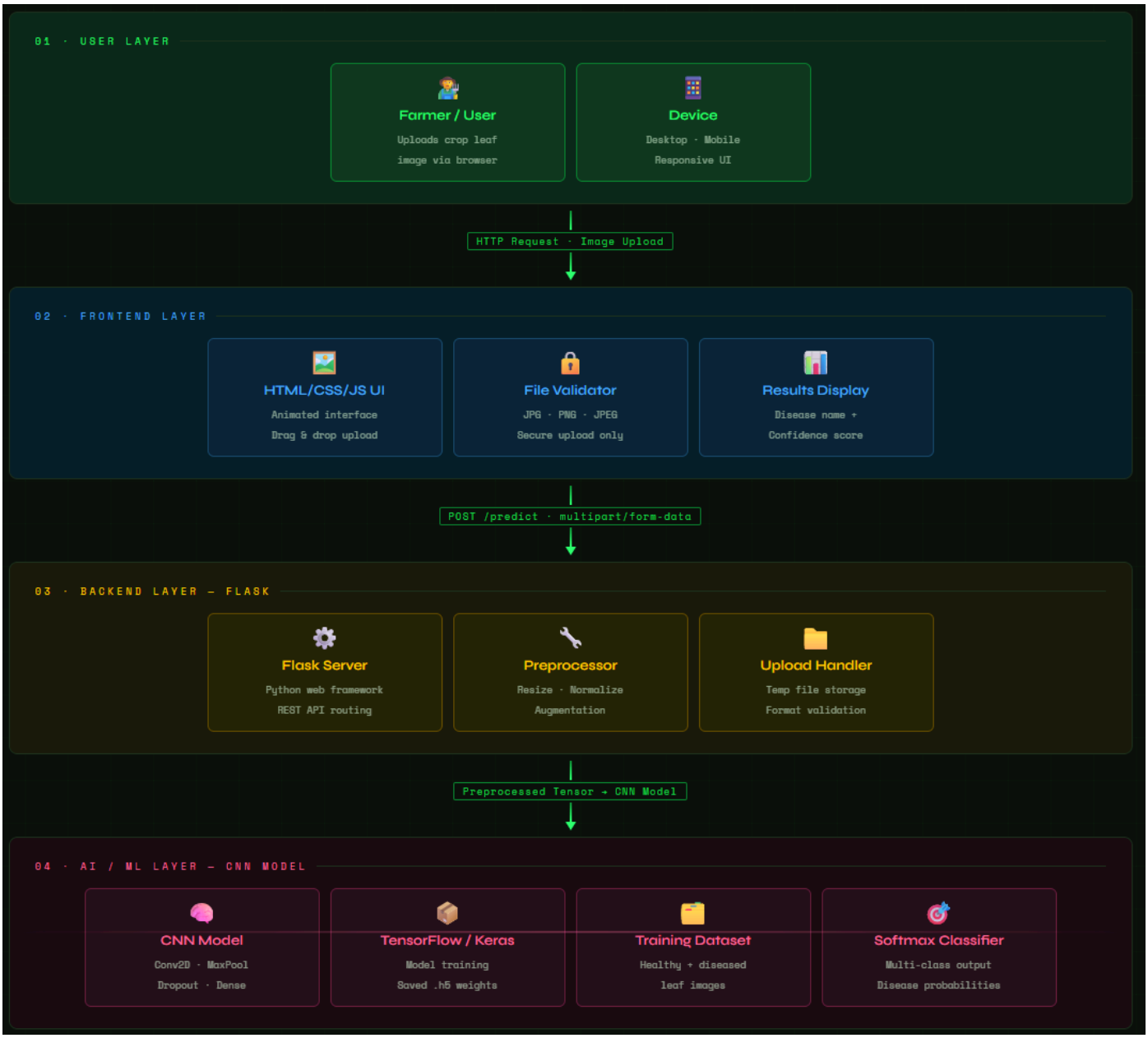

Fig 3.1: Architecture Diagram

4. MODULES DESCRIPTION

Module 1 – Data Collection and Image Preprocessing

This module handles the collection and preparation of crop leaf images for training the CNN model. The dataset contains images of healthy and diseased crop leaves organized into different disease categories. Image preprocessing techniques such as image resizing, rescaling, rotation, zooming, and horizontal flipping are applied using TensorFlow's ImageDataGenerator. These preprocessing and augmentation methods improve model accuracy, reduce overfitting, and increase the diversity of training data. The processed images are then converted into numerical arrays suitable for input to the CNN model.

Module 2 – CNN Model Training and Disease Classification

This module is responsible for training the Convolutional Neural Network (CNN) model using the preprocessed crop leaf images. The CNN model contains convolutional layers, max pooling layers, dense layers, and dropout layers for feature extraction and disease classification. The model learns important image features such as color, texture, and disease patterns from the training dataset. The model is compiled using the Adam optimizer and categorical cross-entropy loss function. Performance metrics such as accuracy, loss, confusion matrix, precision, and recall are used to evaluate the model. After training, the model is saved as `model.keras`, and the disease labels are stored in a JSON file for future prediction.

Module 3 – Flask Backend and Prediction Processing

This module implements the backend server using the Flask web framework. Flask handles image uploads, file validation, prediction processing, and communication between the frontend interface and the trained CNN model. When a user uploads a crop leaf image through the web application, the backend preprocesses the image using the same preprocessing pipeline used during training. The processed image is passed to the loaded CNN model for disease prediction. The backend then receives the predicted disease category and confidence score from the model and sends the result to the frontend for display.

Module 4 – Web Application Interface and Result Display

This module manages the frontend interface of the Crop Disease Detection System using HTML, CSS, and JavaScript. The web application provides a modern and user-friendly interface where users can upload crop leaf images using drag-and-drop or file selection functionality. The uploaded image is previewed on the screen before analysis. After prediction, the system displays the detected disease name, confidence percentage, and uploaded image in real time. The application also provides animated UI components and responsive design features for better user experience across different devices.

5. IMPLEMENTATION AND RESULTS

5.1 Experimental Setup

The Crop Disease Detection System was developed and tested using Python 3.x with TensorFlow and Keras for CNN model training and disease classification. Flask was used as the backend framework, while HTML, CSS, and JavaScript were used to create the frontend web interface. The system was trained using a crop leaf image dataset containing healthy and diseased leaf images. Image preprocessing and augmentation techniques such as resizing, rescaling, rotation, zooming, and horizontal flipping were applied using ImageDataGenerator to improve model performance and accuracy.

The CNN model was trained and evaluated on the processed dataset using performance metrics such as accuracy, precision, recall, confusion matrix, and classification report. Matplotlib and Seaborn libraries were used for visualization of accuracy graphs, loss graphs, and confusion matrix heatmaps. The Flask application was tested locally through a web browser, where users could upload crop images and receive real-time disease prediction results along with confidence scores.

The system was implemented on a standard computer environment with sufficient processing power for model training and testing. Early stopping techniques were applied during training to prevent overfitting and improve model efficiency. The trained model was saved and integrated with the Flask application for real-time prediction. The overall setup provided a reliable and efficient environment for automatic crop disease detection and evaluation.

The experimental setup successfully demonstrated the practical implementation of Deep Learning techniques in smart agriculture for automated crop disease detection. The system

achieved stable and reliable performance during both training and testing phases. The trained CNN model accurately identified different crop diseases from leaf images and generated prediction results with high confidence. The integration of the CNN model with the Flask web application enabled real-time disease prediction through a simple and user-friendly interface. Users were able to upload crop leaf images and receive prediction results quickly and efficiently. The generated evaluation metrics such as accuracy graphs, loss graphs, confusion matrix, and classification reports confirmed the effectiveness of the model in disease classification tasks. The use of image preprocessing and augmentation techniques further improved the model's performance and reduced overfitting. Overall, the experimental setup provided an efficient, scalable, and practical solution for supporting farmers and agricultural users in early crop disease identification and smart farming applications.

5.1 Results

The Crop Disease Detection System was successfully developed and tested using Convolutional Neural Networks (CNN) and Deep Learning techniques. The trained model accurately identified crop diseases from uploaded leaf images and achieved high validation accuracy during evaluation. Image preprocessing and augmentation techniques improved the model's performance and helped reduce overfitting. The generated accuracy graphs, loss graphs, confusion matrix, and classification reports confirmed the effectiveness of the CNN model in disease classification tasks.

The web-based application successfully provided real-time disease prediction through a user-friendly interface developed using Flask, HTML, CSS, and JavaScript. Users were able to upload crop leaf images and receive prediction results along with confidence scores instantly. The system demonstrated fast response time, reliable performance, and efficient image processing, making it suitable for practical applications in smart agriculture and crop disease monitoring.

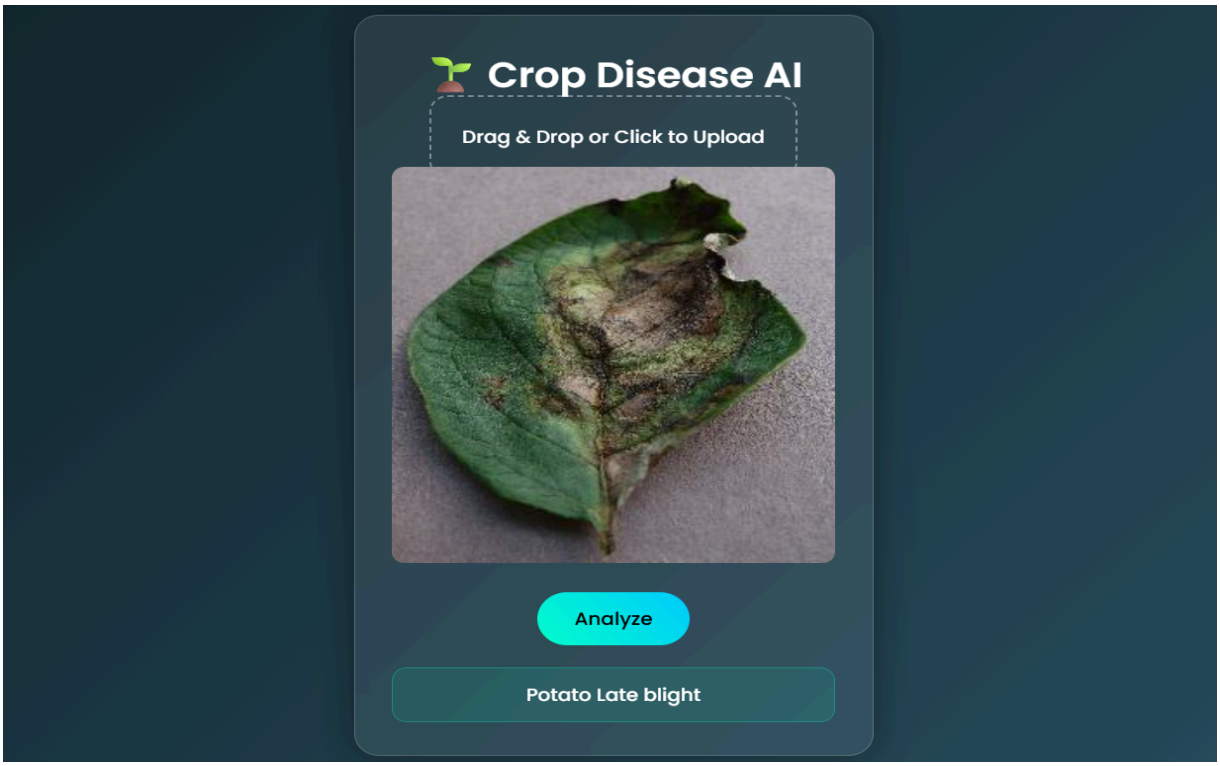

Fig 5.1.a : Predicted-Potato Late blight

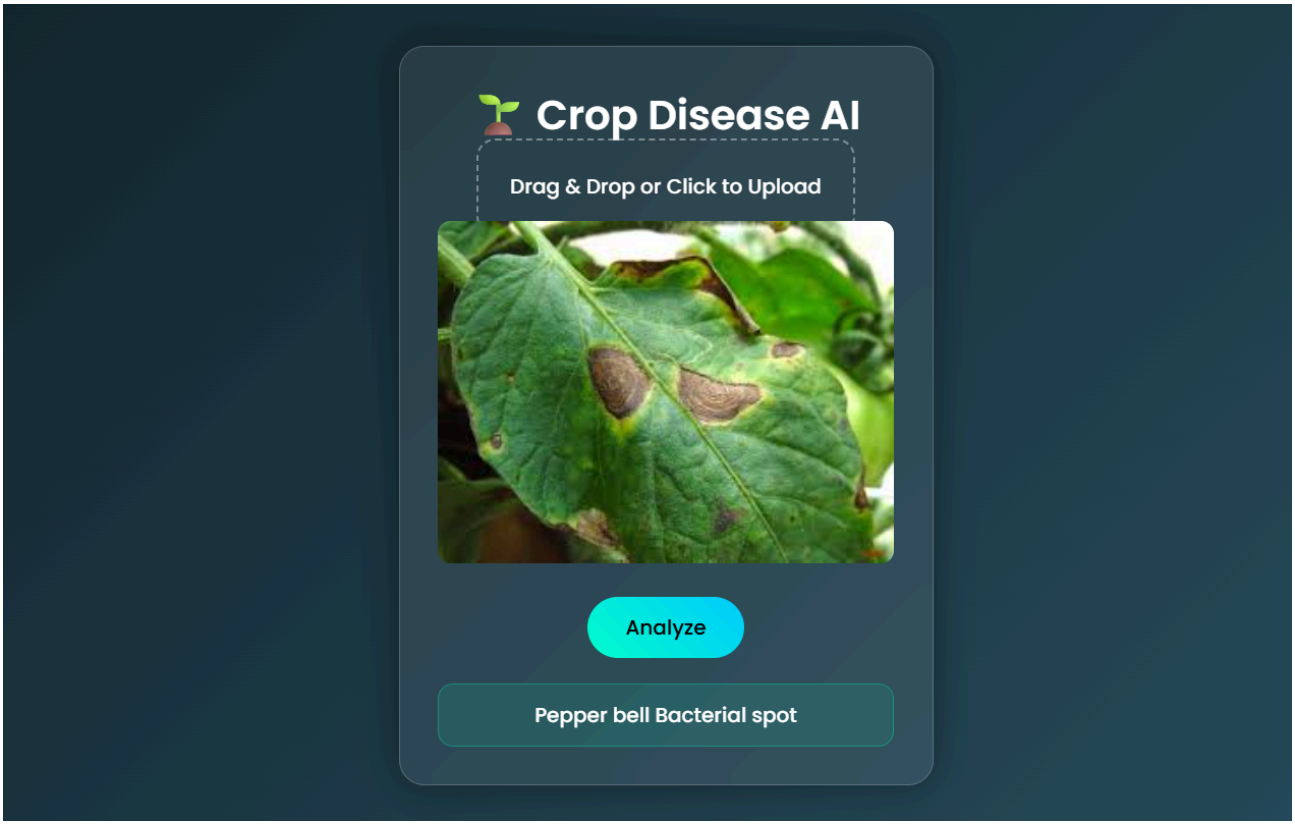

Fig 5.1.a : Predicted-Pepper bell Bacterial Spot

6. CONCLUSION AND FUTURE WORK

The AI-powered Crop Disease Detection System successfully demonstrates the use of Deep Learning and Convolutional Neural Networks (CNN) in smart agriculture. The system accurately identifies crop diseases from leaf images and provides real-time prediction results through a web-based application. TensorFlow, Keras, and Flask were effectively integrated to develop a reliable and user-friendly system for automated disease detection. The generated evaluation metrics such as accuracy graphs, loss graphs, confusion matrix, and classification reports confirmed the good performance of the trained CNN model.

The project provides several advantages for farmers and agricultural users by enabling early disease detection, reducing crop damage, and improving agricultural productivity. The web application allows users to upload crop leaf images easily and receive disease predictions along with confidence scores instantly. Image preprocessing and augmentation techniques improved model performance and helped reduce overfitting during training. The system demonstrated stable performance, fast response time, and efficient image processing capabilities.

In future work, the system can be enhanced using larger crop disease datasets and advanced Deep Learning models such as ResNet or MobileNet to improve prediction accuracy. Mobile application support, cloud deployment, multilingual features, and disease treatment recommendations can also be added to improve usability and accessibility. These future enhancements can make the system more scalable, intelligent, and practical for real-world smart agriculture applications.

7. REFERENCES

1. Chollet, F. (2021) *Deep Learning with Python*. 2nd edn. Manning Publications.
2. Goodfellow, I., Bengio, Y. and Courville, A. (2016) *Deep Learning*. Cambridge, MA: MIT Press.
3. Howard, J. and Gugger, S. (2020) *Deep Learning for Coders with Fastai and PyTorch*. O'Reilly Media.
4. Pedregosa, F. et al. (2011) 'Scikit-learn: Machine Learning in Python', *Journal of Machine Learning Research*, Vol.12, pp.2825–2830.
5. TensorFlow Developers (2023) *TensorFlow Documentation*. Available at: https://www.tensorflow.org/ (Accessed: May 2026).
6. Grinberg, M. (2018) *Flask Web Development: Developing Web Applications with Python*. 2nd edn. O'Reilly Media.
7. Mohanty, S.P., Hughes, D.P. and Salathé, M. (2016) 'Using Deep Learning for Image-Based Plant Disease Detection', *Frontiers in Plant Science*, Vol.7, pp.1–10.